

SECTION AAnswer **all** questions.

1. A laboratory technician needs to make a 230 pF capacitor using two square metal plates of side length 8.7 cm.

(a) Calculate the separation of the metal plates in order to produce this capacitance. [3]

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(b) The technician states that the 230 pF capacitor stores approximately 0.9 μJ of energy when it stores a charge of 20 nC on its plates. Determine whether or not the technician is correct. [3]

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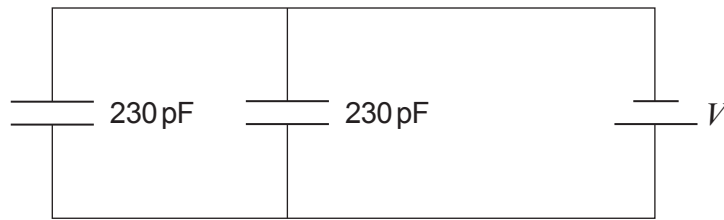
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- (c) By discussing the pd and charge on each of the capacitors, explain why the capacitance of the two 230 pF capacitors connected in parallel is 460 pF. [3]



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- (d) **Draw a diagram** to show how you could combine three capacitors of 230 pF to provide a capacitance of approximately 150 pF. [1]

